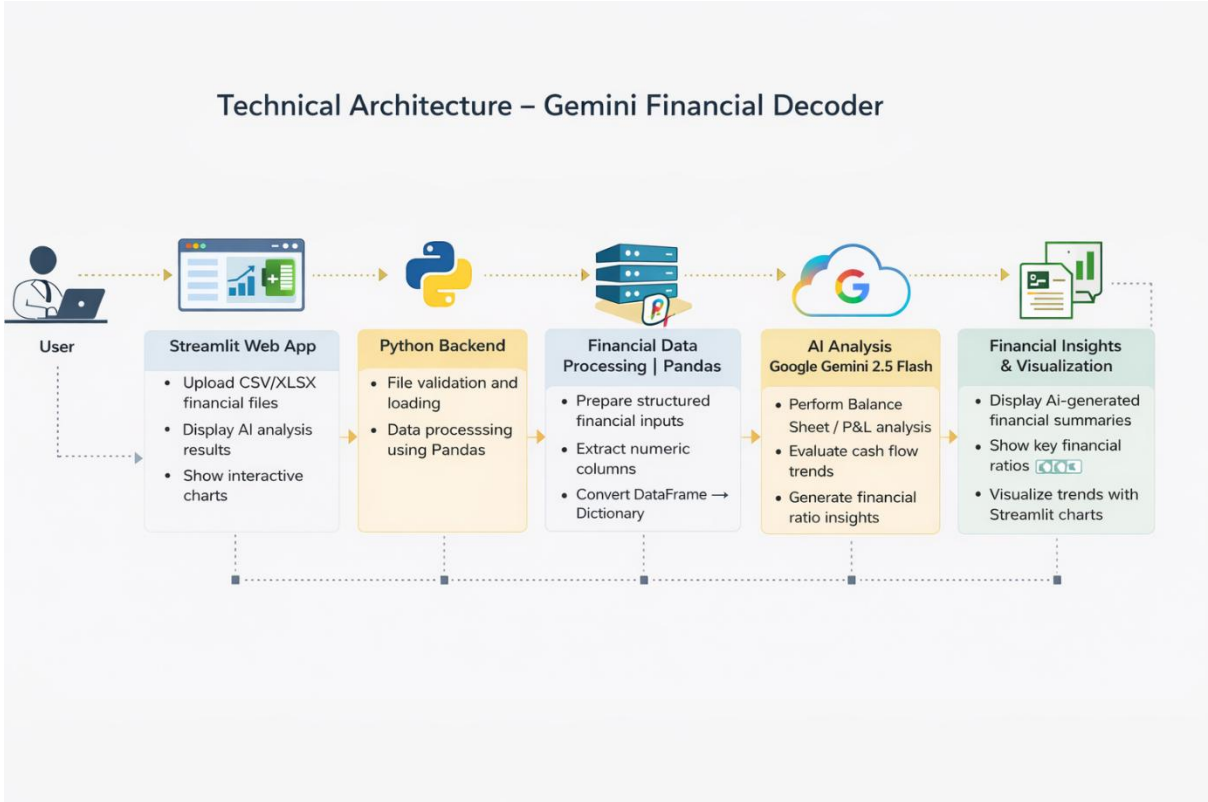
Project Design Phase
Solution Architecture

Date	15 February 2025
Team ID	LTVIP2026TMIDS77563
Project Name	Gemini Financial Decoder
Maximum Marks	4 Marks

Solution Architecture:

The Gemini Financial Decoder follows a layered architecture that connects user interaction, backend processing, AI analysis, and visualization. The system allows users to upload financial statements such as Balance Sheet, Profit & Loss, and Cash Flow data in CSV or Excel format through a Streamlit web interface. The uploaded data is processed using a Python backend where Pandas is used to validate, clean, and structure the financial data.

The processed financial data is then converted into a structured format and sent to the Google Gemini AI model for analysis. The AI model generates financial summaries, interprets key indicators, and provides insights related to financial health, liquidity, leverage, and performance trends.

Finally, the results are displayed to the user through interactive charts and structured summaries within the Streamlit interface. This architecture ensures smooth data flow between the user interface, data processing layer, AI analysis layer, and visualization components, enabling efficient and scalable financial analysis.

Example - Solution Architecture Diagram: